

Vive R07 | OpenXR activation

Software result

R07 enables the installed UE 5.8.2 OpenXR project plugin, packages its loader and requests VR through the review launcher. Build/cook/stage/archive exit 0. A 302-second VR-requested launch and normal window-close exit 0. This is an XR-configured software build; no real XR session was created on this machine.

Required project changes

OpenXR activation brings its installed XRBase dependency. The existing trigger mapping context is registered before XR action creation, without automatic activation in unrelated maps. A new R07-only non-colliding helper requests application Local tracking origin for the already elevated, HMD-locked first-person camera. No floor-height guess, recenter or world-scale change. The review launcher adds -vr and selects R07; original default map is retained.

Controller and movement checks

Vive profile /interaction_profiles/htc/vive_controller binds either hand trigger value/click to per-hand highest-absolute-value Axis1D actions. One horizontal camera-yaw forward request per frame; release stops trigger-only motion. Focus/tracking loss requires release before resuming. All 26 input cases passed: 13 raw-key cases and 13 EnhancedInput action-injection cases matching the installed OpenXR input route. Five keyboard coexistence cases also pass. Existing speed, collision and turning retained; tested speed stays capped at 360 cm/s.

Preservation

All 2,232 R06 actors and existing dependencies match; R07 has 2,233 actors, adding only the origin helper. Nine original protected files remain unchanged; exactly two configuration exceptions are OpenXR activation and trigger-context registration. Rendering, materials, lights/effects, input assets and original maps remain. R06 source and review archive are frozen rollback authorities.

Vive R07 | Runtime and acceptance

Exact runtime blocker

SteamVR was not found in any of the four registered Steam libraries. Standard HKLM and HKCU OpenXR ActiveRuntime values are absent, and XR_RUNTIME_JSON is unset. The packaged OpenXR loader reports extension enumeration failure and no active tracking provider. It cannot establish SteamVR operation without runtime setup. No installation, OS active-runtime selection or security/permission change was performed.

Remaining acceptance

The parent must resolve authorization for SteamVR installation/runtime selection on the intended hardware, then test actual Vive session creation, tracking, trigger press/release, reconnect/focus, camera height, collision, frame rate and comfort. These diagnostic cases simulate head yaw, tracking and focus; they do not prove hardware bindings or stereo rendering. R06 reached 14/16 route probes with two inherited obstructions; full R07/headset traversal is not certified. Cold-start PSO hitches remain disclosed; settings preserved.

Review and publication

C:\Users\mikef\Documents\Codex\2026-09-30\task-3\CampusCenter-

vive\review_build_vive_openxr_r07_r01\Windows\Review_CampusCenter_Vive_R07_OpenXR.cmd

Use this launcher to select R07 and request VR. Raw executable retains the original default map. Use frozen R06 archive/launcher for rollback. Parent owns publication and its readiness hold; no independent push.

Authority and evidence

Installed OpenXR.uplugin and OpenXRInput.cpp define plugin dependencies, Vive profile handling and default mapping-context consumption. Epic SteamVR guidance: <https://dev.epicgames.com/documentation/unreal-engine/developing-for-steamvr-in-unreal-engine?lang=en-US> . final-control-handoff.json pins source/archive/evidence hashes; control-source-manifest.json includes the 12-file source ZIP, binding paths and official references. runtime-validation.json includes actual plugin/loader/registration/failure log evidence.

Matched packaged fallback view

Frozen R06 at left; R07 at right. Same camera and screen percentage 50. R07 requested VR but lacked an installed runtime, so this is a flat-screen fallback comparison, not a stereo/headset test.



Retained visual content

Case glass, materials, supplied awards and hall lighting remain. Full actor/component and dependency checks find no changes to existing visual content. Packaged logs have zero material/shader-usage/fatal failures. OpenXR runtime initialization failure is separately retained as the exact device-readiness blocker.